

1. Bead on a wire: A bead slides without friction down a wire that has the shape $y = f(x)$ in a uniform gravitational field (corresponding to an acceleration g).

(a) Obtain the Hamiltonian of the system.

(b) Also, write down the Hamilton's equation of motion.

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m f'^2 \dot{x}^2 - m g f$$

$$p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} + m f'^2 \dot{x}$$

$$\therefore H = \left(m \dot{x} + m f'^2 \dot{x} \right) \dot{x} - \frac{m \dot{x}^2}{2} - \frac{m f'^2 \dot{x}^2}{2} + m g f$$

$$H = \frac{p^2/2m}{1+f'^2(x)} + m g f(x)$$

$$\frac{\partial H}{\partial x} = \frac{-p^2/2m \cdot 2f' \cdot f''}{(1+f'^2)^2} + m g f'(x) = -\dot{p}$$

$$\dot{p} = \frac{p^2/2m \cdot 2f' \cdot f''}{(1+f'^2)^2} - m g f'(x)$$

$$\frac{\partial H}{\partial p} = \frac{p/m}{1+f'^2(x)} = \dot{x}$$

2. A complicated Lagrangian: Consider a system described by the Lagrangian

$$L = a \dot{x}^2 + b \frac{\dot{y}}{x} + c \dot{x} \dot{y} + f y^2 \dot{x} \dot{z} + g \dot{y} - k \sqrt{x^2 + y^2},$$

where (a, b, c, g, k) are constants.

(a) What is the Hamiltonian of the system?

(b) What are the quantities that are conserved?

$$a) \frac{\partial \mathcal{L}}{\partial \dot{x}} = 2a\dot{x} + c\dot{y} + f y^2 \dot{z} = P_x$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = \frac{b}{x} + c\dot{x} + g = P_y$$

$$\frac{\partial \mathcal{L}}{\partial \dot{z}} = f y^2 \dot{z} = p_z$$

$$H = p_x \dot{x} + p_y \dot{y} + p_z \dot{z} - \mathcal{L}$$

$$= 2a\dot{x}^2 + \cancel{c\dot{x}\dot{y}} + f y^2 \dot{x}\dot{z} + \cancel{\frac{b\dot{y}}{x}} + \cancel{c\dot{y}\dot{z}} + \cancel{g\dot{y}} + \cancel{f y^2 \dot{z}} - a\dot{x}^2 - \cancel{\frac{b\dot{y}}{x}} - \cancel{c\dot{x}\dot{y}} - \cancel{f y^2 \dot{x}\dot{z}} - \cancel{g\dot{y}} + \kappa \sqrt{x^2 + y^2}$$

$$H = a\dot{x}^2 + f y^2 \dot{x}\dot{z} + c\dot{x}\dot{y} + \kappa \sqrt{x^2 + y^2}$$

$$= \dot{x} p_x - a\dot{x}^2 + \kappa \sqrt{x^2 + y^2}$$

$$H = \frac{p_z p_x}{f y^2} - \frac{a p_x^2}{f^2 y^4} + \kappa \sqrt{x^2 + y^2}$$

b). H is independent of $z \Rightarrow p_z$ conserved.

$$\{p_z, H\} = \sum_i \frac{\partial p_z}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial p_z}{\partial p_i} \frac{\partial H}{\partial q_i}$$

$$= - \frac{\partial H}{\partial z} = 0$$

$x \quad y \quad z$
 $p_x \quad p_y \quad p_z$
 E
 $L_x \quad L_y \quad L_z$

H is independent of time, so energy conserved.

$$\{H, H\} = 0$$

$$L_x = y p_z - z p_y$$

$$\{L_x, H\} = \sum_i \frac{\partial L_x}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial L_x}{\partial p_i} \frac{\partial H}{\partial q_i} = z \frac{\partial H}{\partial y} + () \neq 0$$

$$L_y = -x p_z + z p_x$$

$$H = \frac{p_z p_x}{f y^2} - \frac{a p_x^2}{f^2 y^4} + \kappa \sqrt{x^2 + y^2}$$

$$\{L_y, H\} = \sum \frac{\partial L_y}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial L_y}{\partial p_i} \frac{\partial H}{\partial q_i}$$

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